Week 7: Industrial Sensors & Logic

EE0849 Introduction to Robotics - Detailed Notes

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1 Introduction

Industrial robots rarely work in isolation. They interact with their environment through sensors (inputs) and actuators (outputs). Understanding how to interface with industrial I/O systems and write programs that respond to real-world conditions is essential for practical robot applications.

1.1 Why I/O Matters

A robot without I/O is like a machine that cannot see or touch:

- **Inputs** tell the robot about its environment (part present, door closed, temperature OK)
- **Outputs** let the robot affect its environment (gripper close, signal ready, activate welder)

1.2 Chapter Overview

1. Digital I/O fundamentals
2. Industrial sensors
3. Safety systems
4. TP program logic
5. Practical applications

2 Digital I/O Fundamentals

2.1 What is Digital I/O?

Digital I/O deals with signals that have only two states:

State	Voltage	Logic	Typical Use
ON	24V DC	1	Active/True
OFF	0V DC	0	Inactive/False

This binary nature makes digital signals simple, reliable, and noise-resistant compared to analog signals.

2.2 FANUC I/O Signal Types

The FANUC controller organizes I/O into categories:

2.2.1 Digital I/O (DI/DO)

General-purpose signals for user applications:

- **DI[n]**: Digital Inputs - receive signals from sensors, switches
- **DO[n]**: Digital Outputs - control grippers, conveyors, lights

Typical range: DI[1] to DI[8], DO[1] to DO[8] (expandable)

2.2.2 Robot I/O (RI/RO)

Signals specific to robot operation:

- **RI[n]**: Robot Inputs - external start, hold, reset signals
- **RO[n]**: Robot Outputs - robot running, fault, at position signals

These are typically used for cell-level integration.

2.2.3 User Operator Panel (UI/UO)

Signals for the operator interface:

- **UI[n]**: Buttons on operator panel (cycle start, stop)
- **UO[n]**: Indicators for operator (busy, fault, complete)

2.3 Signal Wiring Concepts

2.3.1 Sourcing vs Sinking

Sourcing (PNP): Output provides current when active

$$I_{source} = \frac{V_{supply} - V_{load}}{R_{load}}$$

Sinking (NPN): Output sinks current to ground when active

$$I_{sink} = \frac{V_{load}}{R_{load}}$$

Most FANUC systems use sourcing (PNP) outputs and can accept either type of input.

2.3.2 Contact Types

Normally Open (NO): Open circuit until activated

$$State_{NO} = \begin{cases} OFF & \text{not activated} \\ ON & \text{activated} \end{cases}$$

Normally Closed (NC): Closed circuit until activated

$$State_{NC} = \begin{cases} ON & \text{not activated} \\ OFF & \text{activated} \end{cases}$$

3 Industrial Sensors

3.1 Proximity Sensors

Proximity sensors detect objects without physical contact.

3.1.1 Inductive Proximity Sensors

Principle: Generate an electromagnetic field; metal objects cause eddy currents that dampen the field.

Characteristics:

- Detect metal objects only
- Sensing range: 1-50 mm typical
- Not affected by dust, dirt, oil
- Fast response (~1 ms)

Detection Distance (typical):

Target Material	Relative Distance
Steel	1.0 (reference)
Stainless Steel	0.85
Brass	0.50
Aluminum	0.45
Copper	0.40

3.1.2 Capacitive Proximity Sensors

Principle: Detect changes in capacitance caused by any material.

Characteristics:

- Detect metals, plastics, liquids, powders
- Sensing range: 1-25 mm typical
- Can detect through thin walls
- More susceptible to environmental factors

Applications:

- Level sensing in tanks
- Detecting non-metallic objects
- Counting through containers

3.2 Photoelectric Sensors

Photoelectric sensors use light to detect objects.

3.2.1 Through-Beam (Opposed Mode)

Configuration: Separate emitter and receiver

Operation:

$$Signal = \begin{cases} ON & \text{beam blocked} \\ OFF & \text{beam clear} \end{cases}$$

Characteristics:

- Longest sensing range (up to 100 m)
- Most reliable detection
- Requires alignment between emitter and receiver

3.2.2 Retro-Reflective

Configuration: Emitter and receiver in same housing, uses reflector

Operation: Light bounces off reflector; object blocks return beam

Characteristics:

- Range up to 10 m
- Easier wiring (one cable)
- Some objects may be too reflective

3.2.3 Diffuse (Proximity Mode)

Configuration: Emitter and receiver in same housing

Operation: Detects light reflected from target object

Characteristics:

- Range up to 2 m (surface dependent)
- Simplest installation
- Background suppression variants available

3.3 Mechanical Sensors

3.3.1 Limit Switches

Types:

- Plunger: Linear actuation
- Lever: Angular actuation
- Roller: Rolling contact

Characteristics:

- Positive mechanical feedback

- High current capacity
- Require physical contact

3.3.2 Microswitches

Smaller versions of limit switches with:

- Precise actuation point
- Snap-action mechanism
- Multiple mounting options

3.4 Sensor Selection Criteria

Factor	Consideration
Target material	Metal → Inductive; Any → Capacitive/Photoelectric
Distance	Long → Through-beam; Short → Proximity
Environment	Dusty → Inductive; Clean → Photoelectric
Speed	Fast → Photoelectric; Slow → Mechanical
Reliability	Critical → Mechanical or through-beam

4 Safety Systems

4.1 Industrial Robot Hazards

Robots present unique hazards:

1. **Impact**: Robot arm collision with personnel
2. **Crushing/Trapping**: Between robot and fixtures
3. **Unexpected motion**: Movement during programming or maintenance
4. **Ejected parts**: Flying debris from grippers or processes

4.2 Fundamental Safety Devices

4.2.1 Emergency Stop (E-Stop)

The E-Stop is the most critical safety device:

Characteristics:

- Red mushroom-head button
- Latching (stays pressed)
- Requires deliberate reset
- Hardwired (not software-dependent)

Location requirements:

- Robot cell perimeter
- Teach pendant
- Operator station
- Inside safety fencing (if accessible)

4.2.2 Deadman Switch (3-Position Enable)

Found on teach pendants:

Position	Action
Released	Motion disabled
Mid-position (held)	Motion enabled
Squeezed (panic)	Motion disabled

This design prevents both "letting go" and "grabbing tighter" panic responses from causing motion.

4.2.3 Light Curtains

Operation: Infrared light beams detect intrusion into protected zone

Parameters:

- Resolution: Distance between beams (14mm, 25mm, 40mm)
- Height: Vertical coverage
- Response time: Time to trigger safety circuit

Selection:

Resolution	Application
14 mm	Finger detection
25 mm	Hand detection
40 mm	Arm/body detection

4.2.4 Safety Mats

Pressure-sensitive floor mats that detect presence:

Characteristics:

- Large coverage area possible
- Self-monitoring for damage
- Can be custom shaped

4.3 Fail-Safe Design Principles

4.3.1 Series Safety Circuit

All safety devices connect in series:

$$Circuit = E\text{-}Stop_1 \land E\text{-}Stop_2 \land LightCurtain \land SafetyMat \land ...$$

If ANY device opens → Circuit opens → Robot stops

4.3.2 Normally Closed Contacts

Safety devices use NC contacts:

Why NC?

Failure Mode	NO Contact	NC Contact
Wire breaks	Appears ON (dangerous!)	Appears OFF (safe)
Contact fails	May appear ON	Appears OFF
Power loss	No signal	Circuit opens

4.3.3 Dual-Channel Safety Relays

Safety relays monitor two independent channels:

- Both channels must agree for operation
- Cross-monitoring detects faults
- Provides Safe Torque Off (STO) to drives

4.4 Safety Standards

Key standards for industrial robot safety:

Standard	Title
ISO 10218-1/2	Robots and robotic devices - Safety
ANSI/RIA R15.06	Industrial robot safety
ISO 13849	Safety-related parts of control systems
IEC 62443	Industrial automation security

5 TP Program Logic

5.1 Program Structure

A FANUC TP (Teach Pendant) program has this structure:

```
/PROG program_name
/ATTR
  (program attributes)
/MN
  1: instruction_1
  2: instruction_2
  ...
/END
```

5.2 Motion Instructions

5.2.1 Motion Types

Type	Command	Path	Use
Joint	J	Curved (fastest)	Point-to-point
Linear	L	Straight line	Process paths
Circular	C	Arc	Curved paths

5.2.2 Motion Syntax

```
J P[1] 100% FINE
            Termination type
          Speed (% or mm/sec)
        Position register
        Motion type
```

5.2.3 Termination Types

FINE: Robot stops at the point

- Velocity reaches zero
- Maximum position accuracy
- Used at pick/place points

CNT (Continuous): Robot blends through the point

$$\text{CNT}n : \text{Blend amount, } n = 0 \text{ to } 100$$

- CNT0: Minimal blend (near-stop)
- CNT50: Medium blend
- CNT100: Maximum blend (fastest)

The blend zone is calculated as:

$$r_{blend} = \frac{n}{100} \times r_{max}$$

where r_{max} depends on speed and path geometry.

5.3 I/O Instructions

5.3.1 Output Control

```
DO[n] = ON          ; Set output ON
DO[n] = OFF         ; Set output OFF
DO[n] = PULSE, t    ; Pulse for t seconds
```

5.3.2 Input Checking

```
IF DI[n]=ON, action     ; Conditional
WAIT DI[n]=ON           ; Wait for signal
WAIT DI[n]=ON + TIMEOUT, LBL[x]  ; Wait with timeout
```

5.4 Control Flow

5.4.1 Labels and Jumps

```
LBL[n]              ; Define label n (1-32767)
JMP LBL[n]          ; Unconditional jump to label
```

5.4.2 Conditional Statements

```
IF condition, action
IF R[1] > 5, JMP LBL[10]
IF DI[1]=ON, DO[2]=ON
```

5.4.3 Loops

FOR Loop:

```
FOR R[1] = 1 TO 10
  ; loop body
ENDFOR
```

WAIT Loop (pseudo-loop):

```
LBL[1]
  ; loop body
  IF condition, JMP LBL[99]
  JMP LBL[1]
LBL[99]
```

5.5 Registers

5.5.1 Numeric Registers R[n]

Store integers or floating-point values:

```
R[1] = 0               ; Initialize
R[1] = R[1] + 1        ; Increment
R[2] = R[1] * 3.14     ; Arithmetic
```

5.5.2 Position Registers PR[n]

Store complete positions:

```
PR[1] = LPOS           ; Current position
PR[1,1] = 100.0        ; Set X component
```

5.6 Wait and Timing

5.6.1 Time Delays

```
WAIT 1.5(sec)          ; Wait 1.5 seconds
```

5.6.2 Signal Waiting

```
WAIT DI[1]=ON          ; Wait indefinitely
WAIT DI[1]=ON + TIMEOUT, LBL[99]  ; 60s default timeout
WAIT DI[1]=ON, TIMEOUT=30sec      ; Custom timeout
```

6 Practical Applications

6.1 Pick and Place with I/O

6.1.1 Basic Sequence

1. Wait for part ready signal (DI)
2. Move to pick position
3. Close gripper (DO)
4. Wait for grip confirmation
5. Move to place position
6. Open gripper (DO)
7. Signal cycle complete (DO pulse)
8. Return and repeat

6.1.2 Grip Settling Time

Always include a WAIT after gripper commands:

```
DO[1]=ON            ; Close gripper
WAIT 0.3(sec)       ; Settling time
```

This allows:

- Pneumatic gripper to fully close
- Part to settle in gripper
- Confirmation signal to stabilize

6.2 I/O Handshaking

For reliable communication with external devices:

6.2.1 Robot Signals to External

```
DO[1]=ON           ; Robot ready
DO[2]=PULSE,0.2sec ; Cycle complete
DO[3]=ON           ; Fault indication
```

6.2.2 External Signals to Robot

```
WAIT DI[1]=ON      ; Part in position
IF DI[2]=ON, ...   ; External ready
WAIT DI[3]=OFF     ; Obstruction cleared
```

6.2.3 Handshake Protocol

1. Robot: Sets READY signal (DO[1]=ON)
2. External: Responds with PART_READY (DI[1]=ON)
3. Robot: Clears READY, performs operation
4. Robot: Sets COMPLETE signal (DO[2]=PULSE)
5. External: Clears PART_READY
6. Repeat

6.3 Error Handling Patterns

6.3.1 Timeout Handling

```
WAIT DI[1]=ON + TIMEOUT, LBL[90]
; ... normal operation ...
JMP LBL[99]

LBL[90]        ; Timeout handler
DO[10]=ON      ; Fault indicator
PAUSE          ; Wait for operator
DO[10]=OFF
JMP LBL[1]     ; Retry

LBL[99]        ; Normal exit
```

6.3.2 Retry Logic

```
R[10] = 0      ; Retry counter
LBL[1]
R[10] = R[10] + 1
IF R[10] > 3, JMP LBL[90]  ; Max retries exceeded
; ... operation ...
IF DI[5]=OFF, JMP LBL[1]   ; Retry if failed
; ... continue ...
```

7 Summary

7.1 Key Concepts

1. **Digital I/O**: Binary signals for robot-environment interaction

2. **Sensors**: Proximity (inductive, capacitive), photoelectric, mechanical
3. **Safety**: E-stop, deadman switch, light curtains - all fail-safe
4. **TP Programs**: J/L/C motion, FINE/CNT termination, IF/WAIT logic
5. **Handshaking**: Coordinated signal exchange for reliable operation

7.2 Best Practices

7.2.1 I/O Design

- Use NC contacts for safety devices
- Include settling time after gripper operations
- Implement timeouts on all WAIT commands
- Use handshaking for external coordination

7.2.2 Programming

- Test in single-step mode first
- Use FINE at pick/place positions
- Use CNT for travel motions
- Add error handling and retry logic

7.2.3 Safety

- Never bypass safety devices
- Test safety systems regularly
- Use dual-channel safety circuits
- Follow applicable safety standards

7.3 Common Mistakes

1. No settling time after gripper operation
2. Using CNT at precision positions
3. Missing timeout on WAIT commands
4. Safety wiring with NO instead of NC contacts
5. Single-channel safety circuits